

CAT 2025

MBA FASTRACK



Lecture – 2

Geometry 2

Triangles and Circles

By- RAHUL BATHLA



 $\rightarrow$ General PROPERTIES

$\rightarrow$ RIGHT / EQUI.

$\rightarrow$ SPECIAL CENTERS / SPECIAL LINES

$\rightarrow$ SIMILARITY / CONGRUENCY

GEOMETRY found.
MBA WALLAH

$\left\{ \begin{array}{l} * \text{ SURFACE AREA} \\ * \text{ QUAD} \\ * \text{ POLYGON} \end{array} \right\}$

CIRCLES

TOPICS *to be covered*



Topic

Important Lines and Centers

Topic

Congruency + Similarity

Topic

Circles

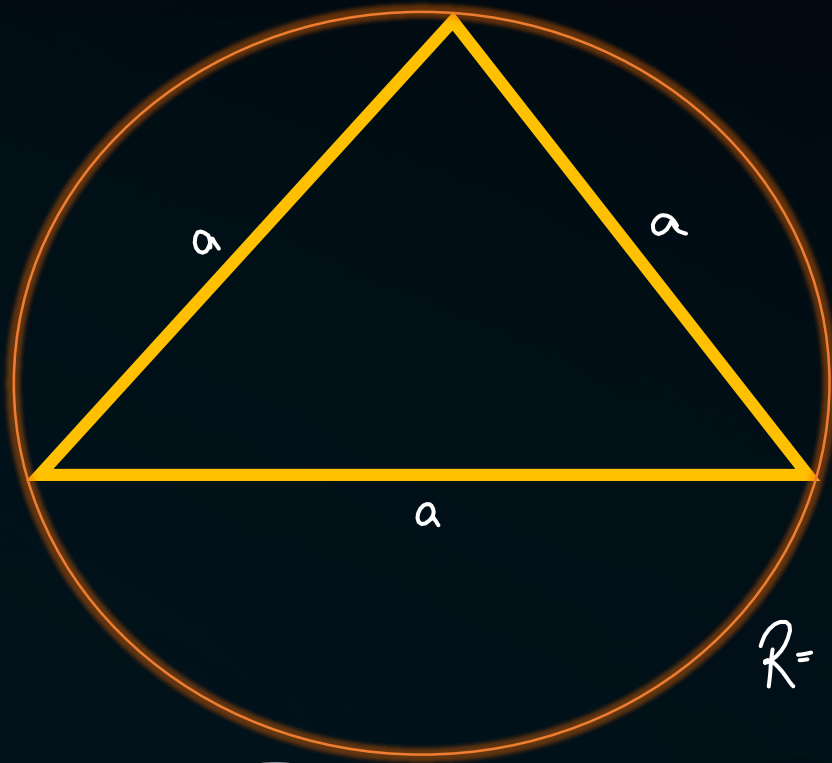


TRUE FOR ALL ΔS

$$R = \frac{abc}{4\Delta}$$

↳ AREA OF TRIANGLE

Circumradius and Two Special Triangles



$$R = \frac{abc}{4\Delta}$$

$$= \frac{a^3}{4 \times \frac{\sqrt{3}}{4} a^2}$$

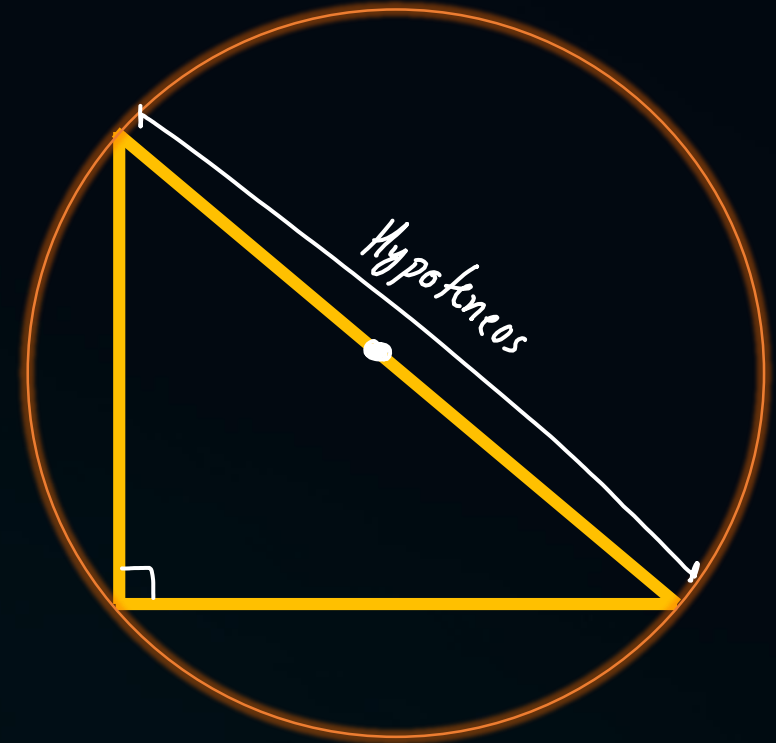
$$= \frac{a}{\sqrt{3}}$$

$$r = \frac{a}{2\sqrt{3}}$$

$$R = \frac{a}{\sqrt{3}}$$

$$2r = R$$

$$R:r = 2:1$$

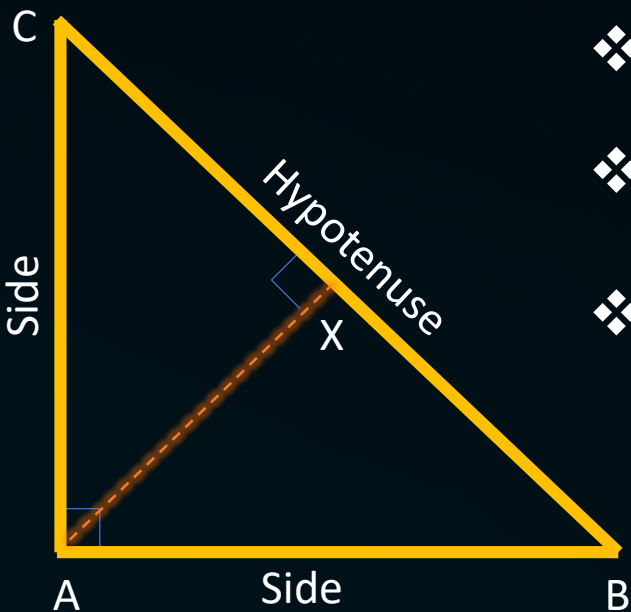


$$R = \frac{\text{hypokneus}}{2}$$

$$r = s - h$$

$$r = \frac{a+b-c}{2}$$

Right Triangle



$$\diamond r = s - \text{hypotenuse}$$

$$\diamond r = \frac{a+b-c}{2}$$

$$\diamond R = \frac{\text{Hypotenuse}}{2}$$

$$\diamond \text{Area} = \frac{1}{2} (\text{Product of Sides})$$

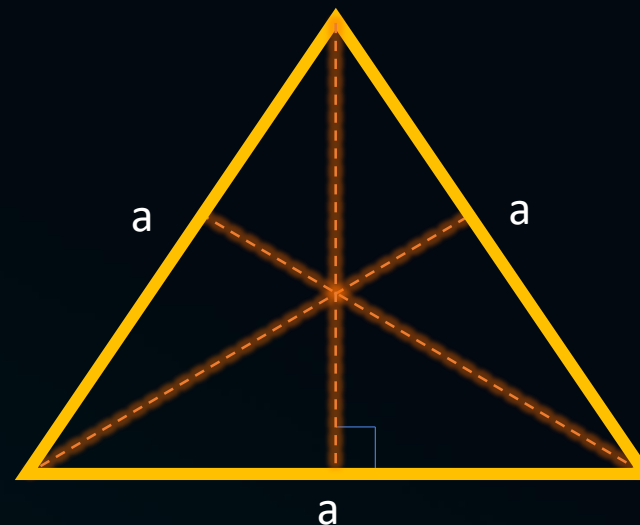
$$\diamond \frac{1}{2} (\text{Product of Sides}) = \frac{1}{2} (\text{hypotenuse})(AX)$$

$$\diamond \text{Pythagoras Theorem}$$

$$3k, 4k, 5k \quad ; \quad 5k, 12k, 13k \quad ; \quad 7k, 24k, 25k$$

$$9k, 40k, 41k \quad ; \quad 16k, 63k, 65k$$

Equilateral Triangle



$$\diamond r = \frac{a}{2\sqrt{3}}$$

$$\diamond R = \frac{a}{\sqrt{3}}$$

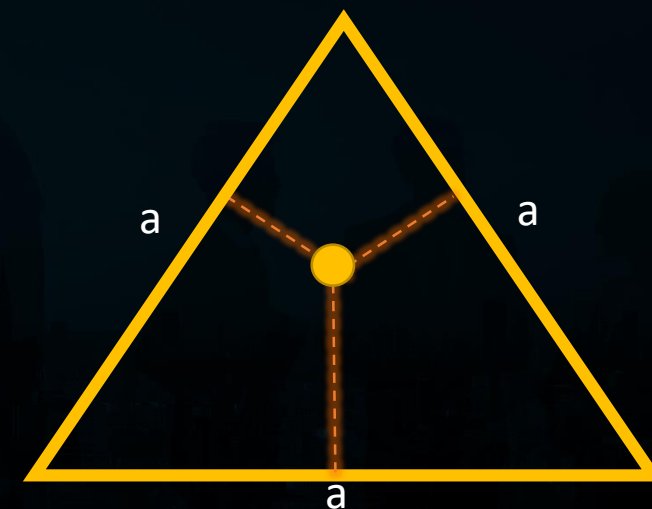
$$\boxed{R:r = 2:1}$$

$$\diamond \text{Area} = \frac{\sqrt{3}}{4} a^2$$

$$\diamond h = \frac{\sqrt{3}}{2} a$$

$$\diamond \text{Area} = \frac{p^2}{\sqrt{3}}$$

$$\text{if } p = a + b + c$$



QUESTION – 1



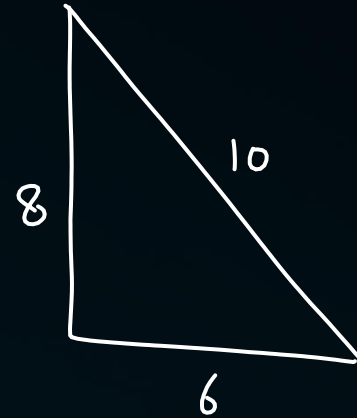
#Q. Find the Circumradius of an Right Triangle whose shorter sides are 6 cm and 8 cm.

A 5 cm ✓✓

B 6 cm

C 3 cm

D 4 cm



QUESTION – 2



#Q. Find the Circumradius of an Equilateral Triangle whose Area is $9\sqrt{3}$.

- A** $\sqrt{3}$
- B** $6\sqrt{3}$
- C** $2\sqrt{3}$ ✓✓
- D** $3\sqrt{3}$

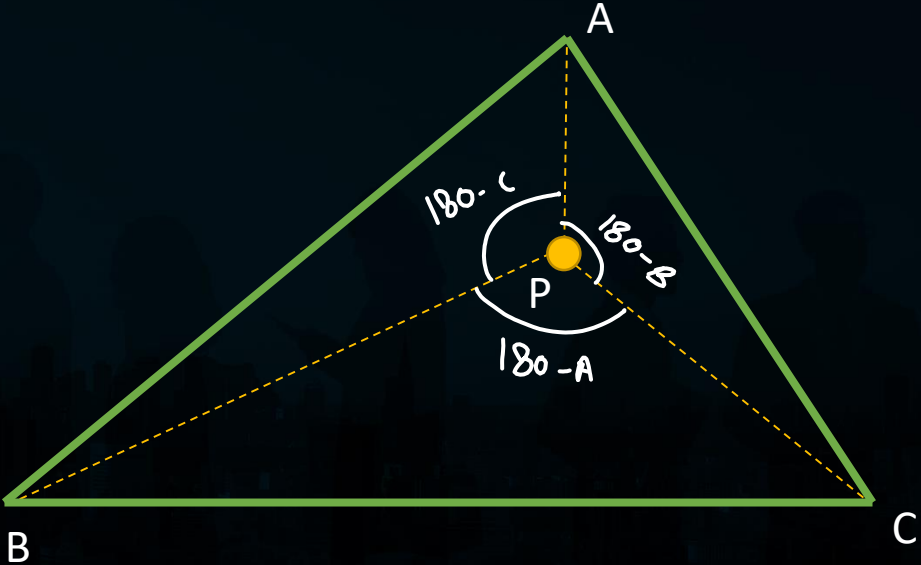
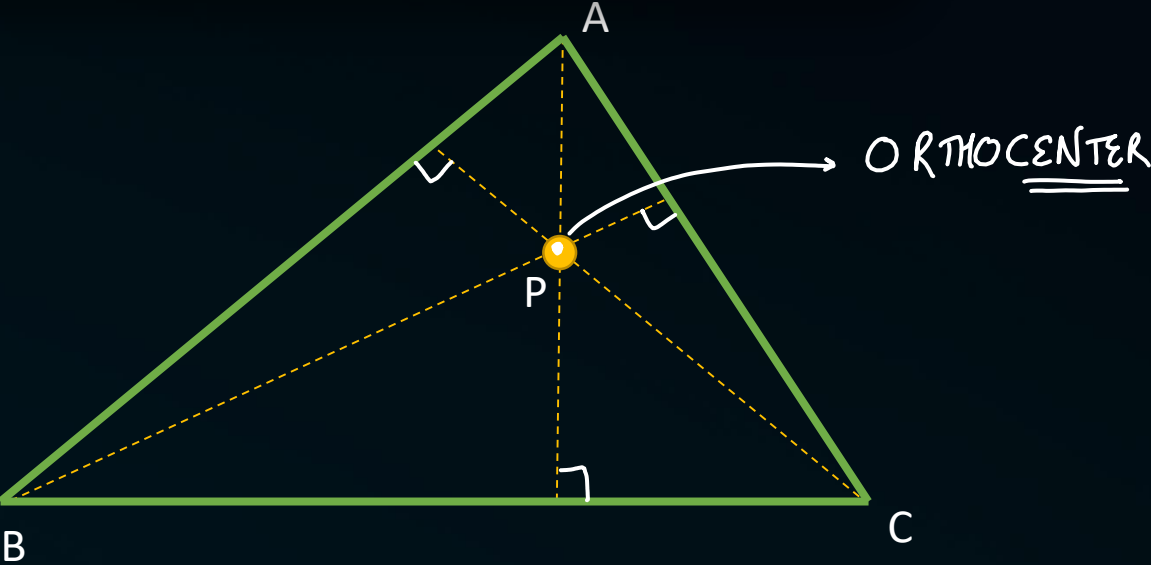
$$\frac{\sqrt{3}}{4} a^2 = 9\sqrt{3}$$

$$a^2 = 36$$

$$\boxed{a = 6}$$

$$R = \frac{a}{\sqrt{3}} = \frac{6}{\sqrt{3}}$$

Orthocentre and Perpendiculars





Topics : Congruency and Similarity

1. Congruent Triangles



- Two Triangles can be proved congruent by following methods.

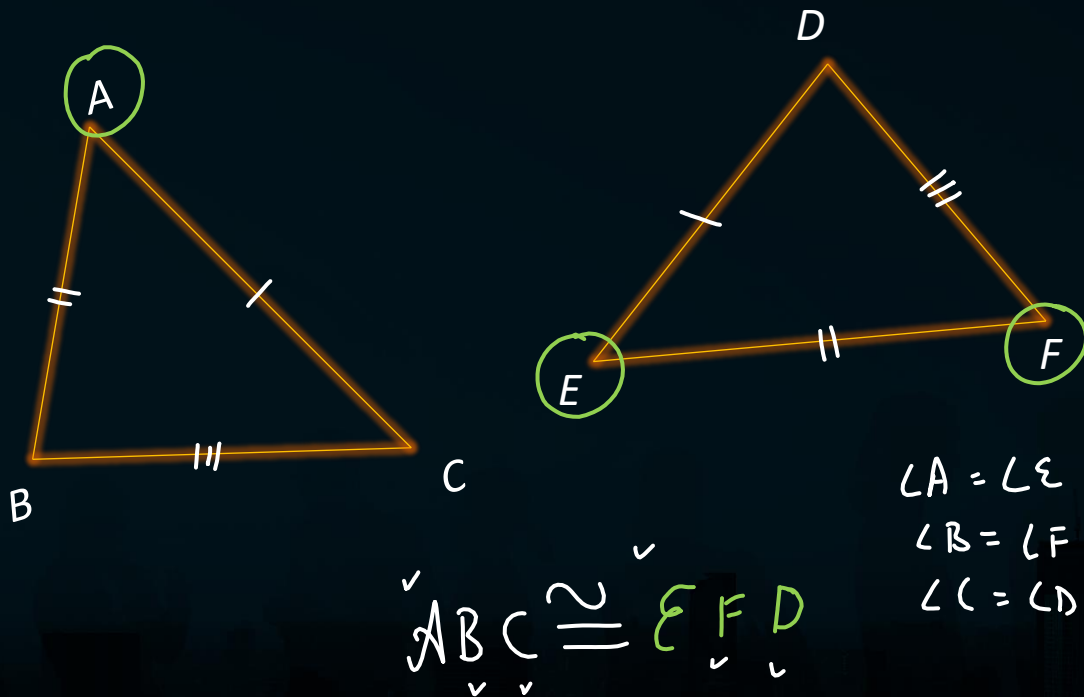
□ SSS

□ ASA

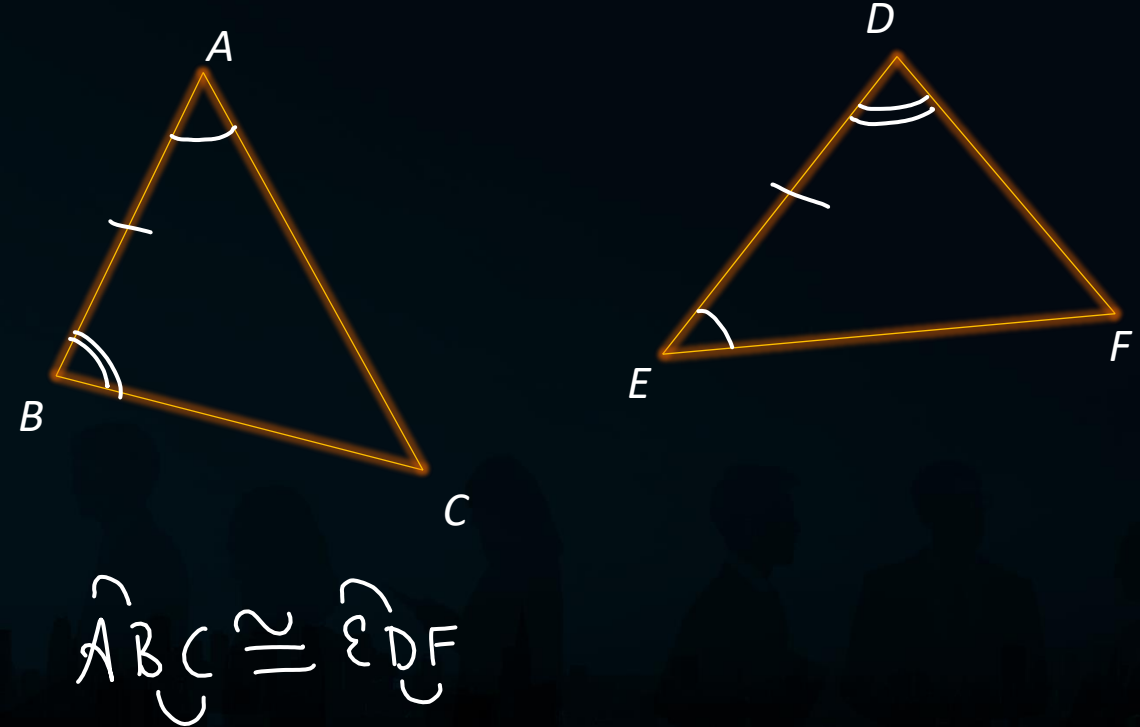
□ SAS

□ RHS

□ SSS

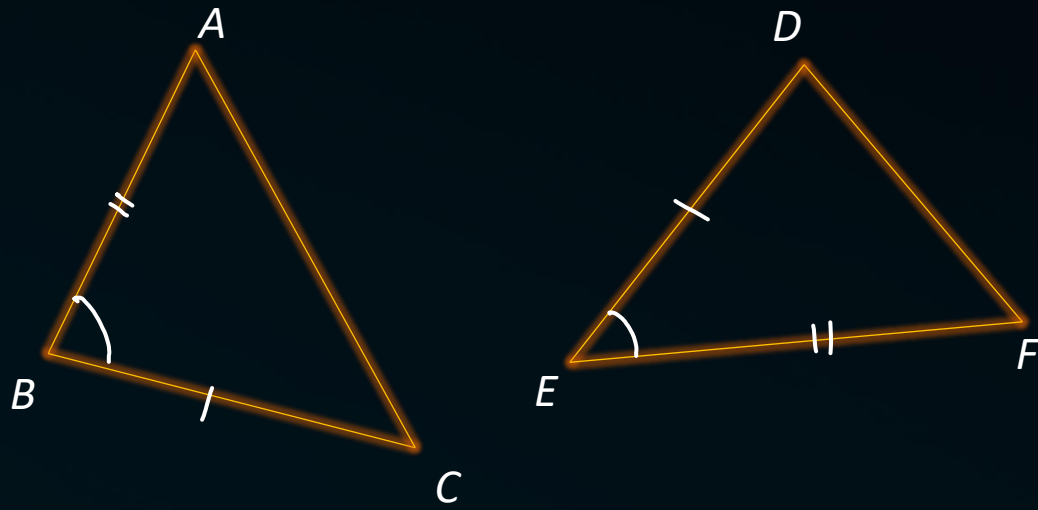


□ ASA



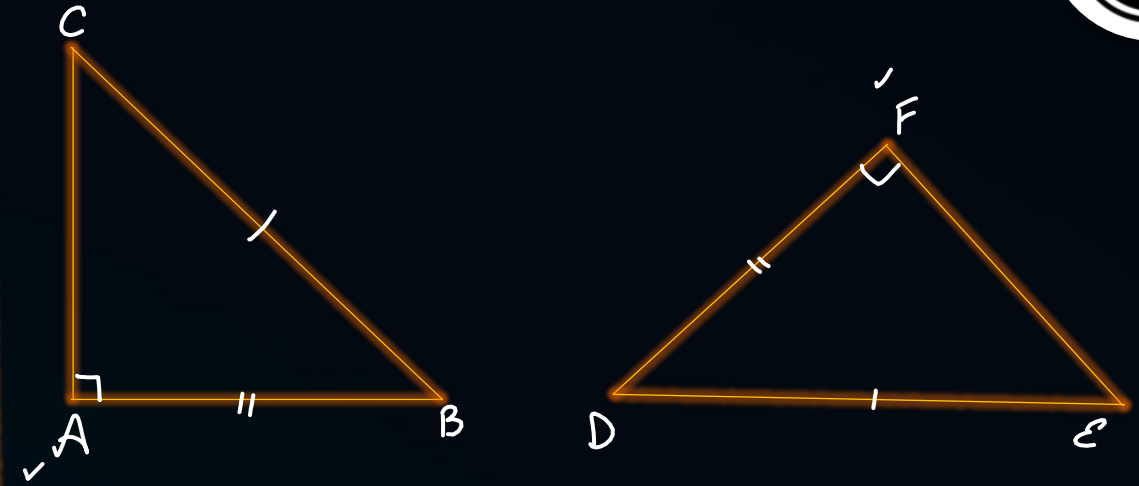
- The Key is to write down the correct corresponding letters at corresponding places

□ SAS



$$\triangle ABC \cong \triangle FED$$

□ RHS



$$\triangle ABC \cong \triangle FED$$



2. Similar Triangles



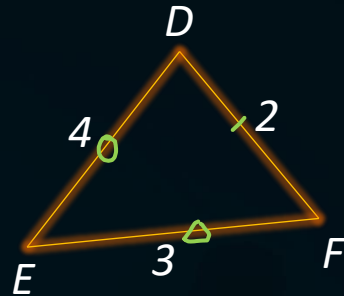
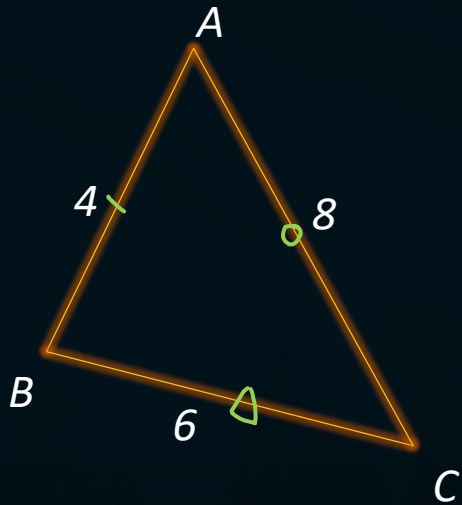
- Two Triangles can be proved similar by following methods.

□ SSS

□ SAS

□ AAA

- If all the three sides of one Triangle are equal to the sides of the other Triangle, The triangles are said to be similar by SSS similarity condition.

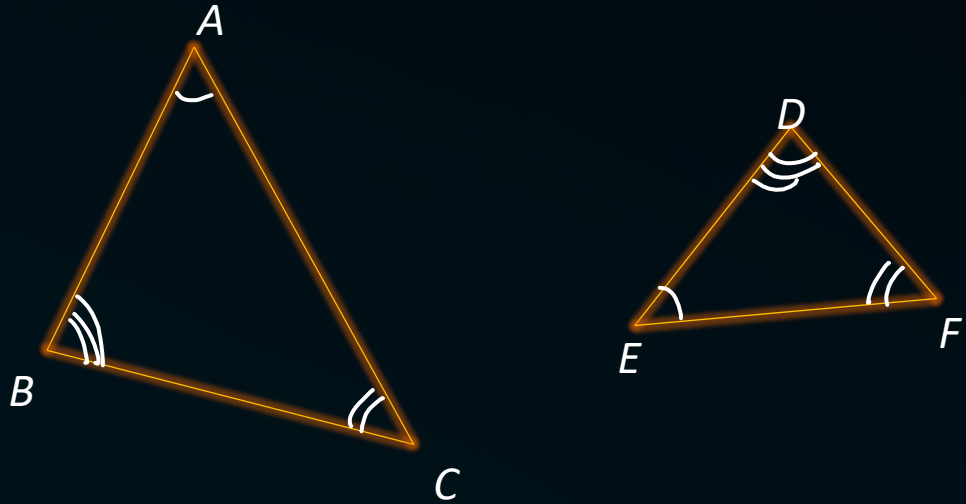


$$\frac{AB}{DE} = \frac{BC}{EF} = \frac{AC}{DF}$$

$$ABC \sim DEF \quad \angle A = \angle D \quad \angle C = \angle F$$

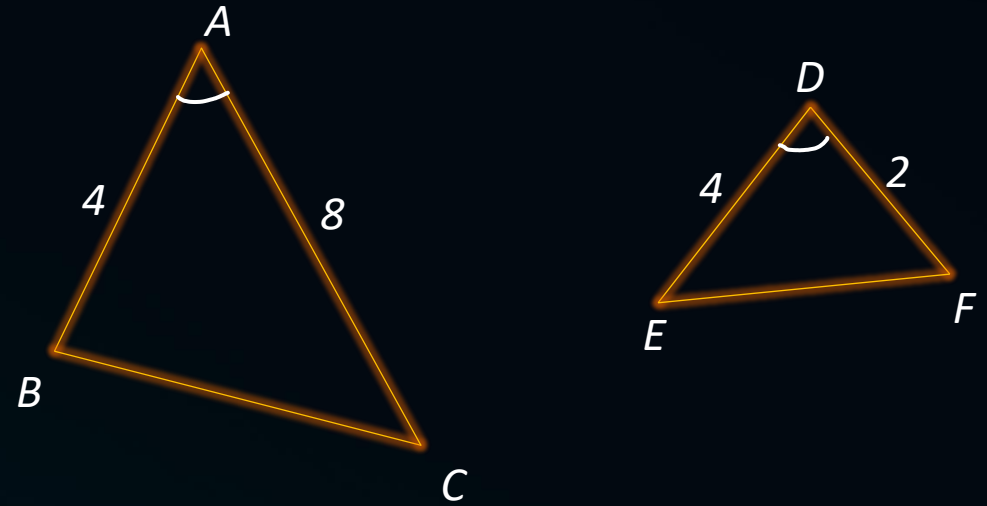
- The Key is to write down the correct corresponding letters at corresponding places

□ AAA



$ABC \sim EDF$

□ SAS



$ABC \sim DEF$



QUESTION - 3



#Q. 11 Find the value of x ? Given that $AB \parallel CD$

A

6 cm

B

1 cm ✓✓

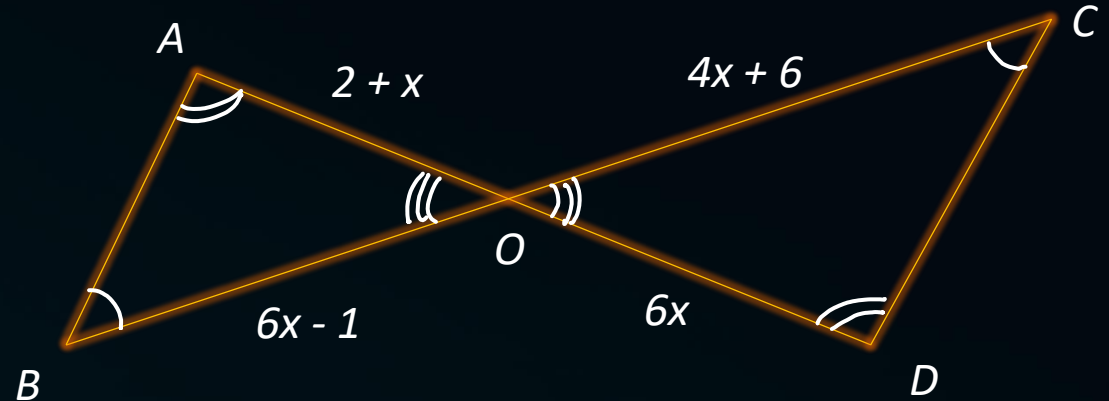
C

3 cm

D

4 cm

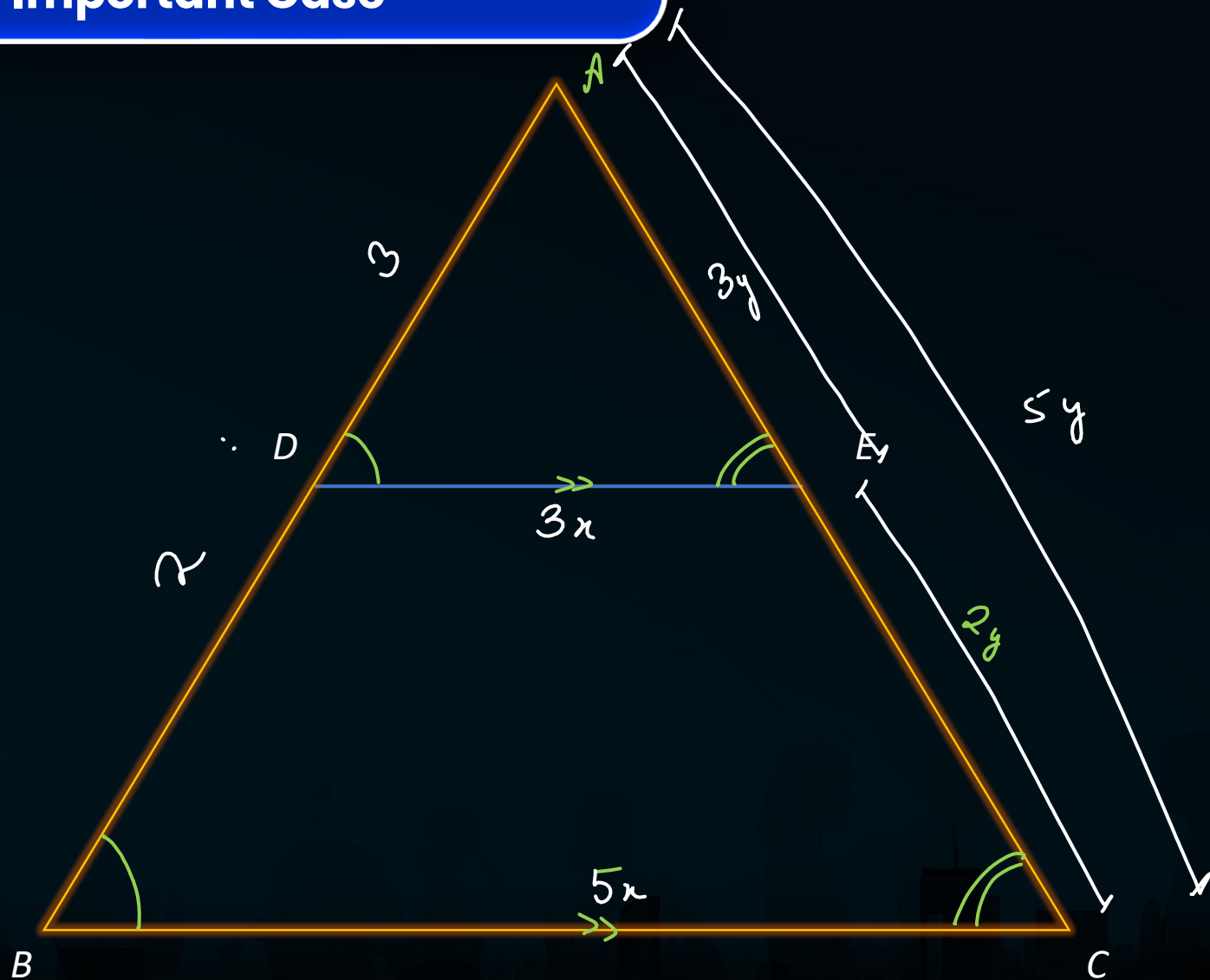
$$\frac{2+x}{6x} = \frac{6x-1}{4x+6}$$



$$\triangle AOB \sim \triangle DOC$$

$$\frac{AO}{DO} = \frac{BO}{CO}$$

Important Case



$$ADE \sim ABC$$

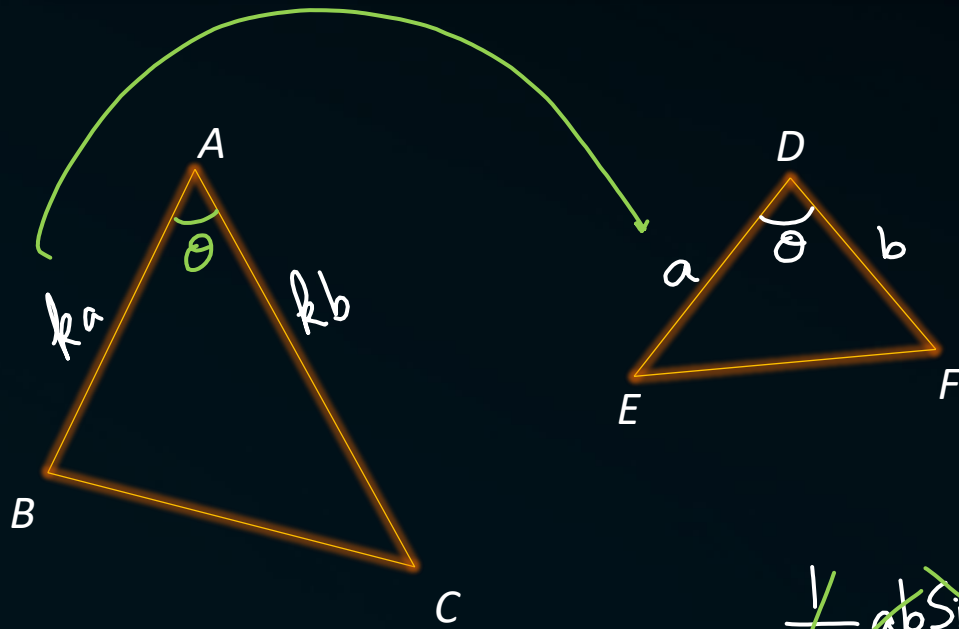
$$\frac{AD}{AB} = \frac{AE}{AC} = \frac{DE}{BC}$$

$$\text{If } \frac{DE}{BC} = \frac{3}{5}$$

$$\text{then } \frac{AE}{AC} = \frac{3}{5}$$

$$\frac{AE}{EC} = \frac{3}{2}$$

Area and Similar Triangles



$$\frac{1}{2} ab \sin \theta$$

$$\frac{\Delta ABC}{\Delta DEF} = \left(\frac{AB}{DE} \right)^2 = \left(\frac{AC}{DF} \right)^2 = \left(\frac{BC}{EF} \right)^2$$

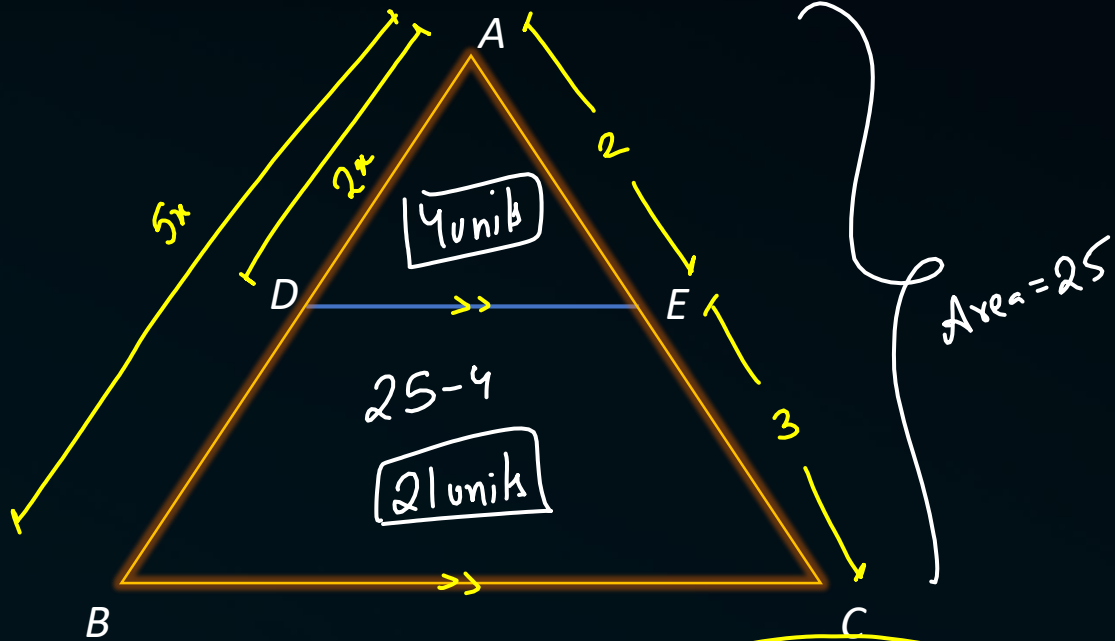
$$\frac{1}{2} ab \sin \theta$$

$$\frac{1}{2} k^2 ab \sin \theta$$

$$k^2 : 1$$

Important Case

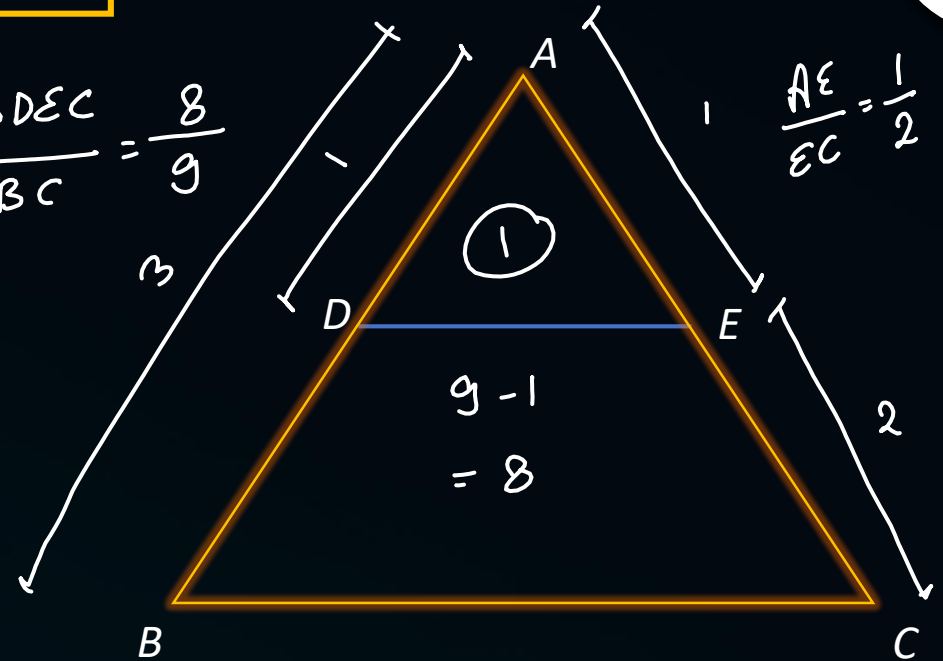
DE // BC



$$\frac{ADE}{ABC} = \frac{4}{25}$$

$$\frac{AD}{AB} = \frac{2}{5}$$

$$\frac{\text{TRAP BDEC}}{\triangle ABC} = \frac{8}{9}$$

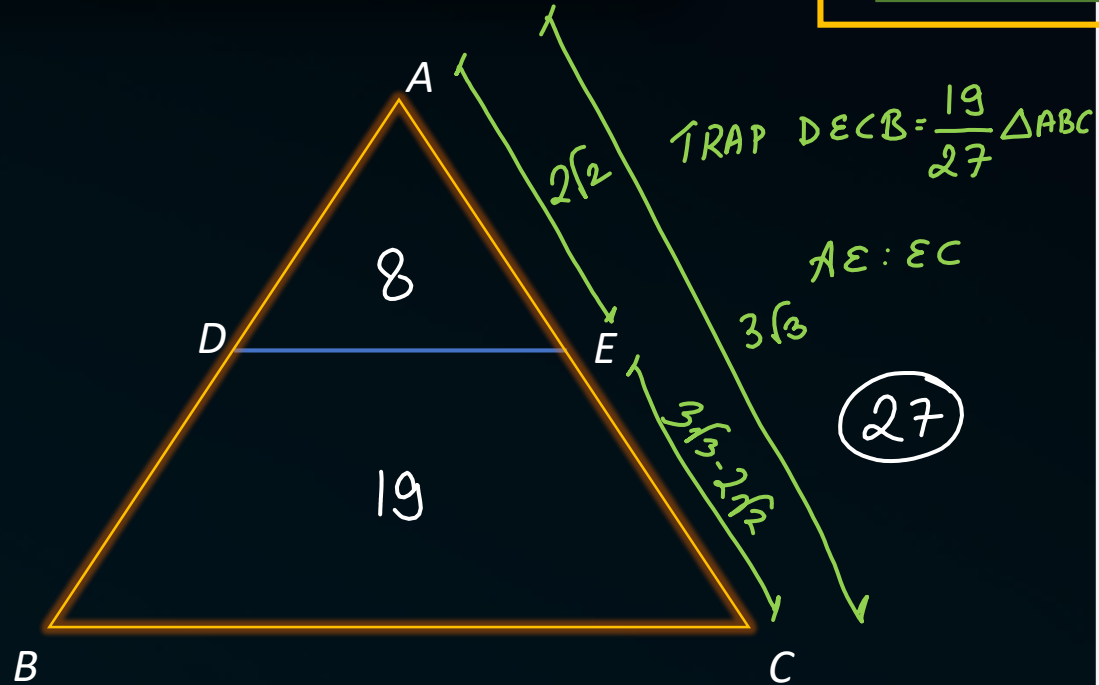


$$\triangle ABC \rightarrow 9$$

$$\triangle ADE = 1$$

YOUR TURN

DE // BC



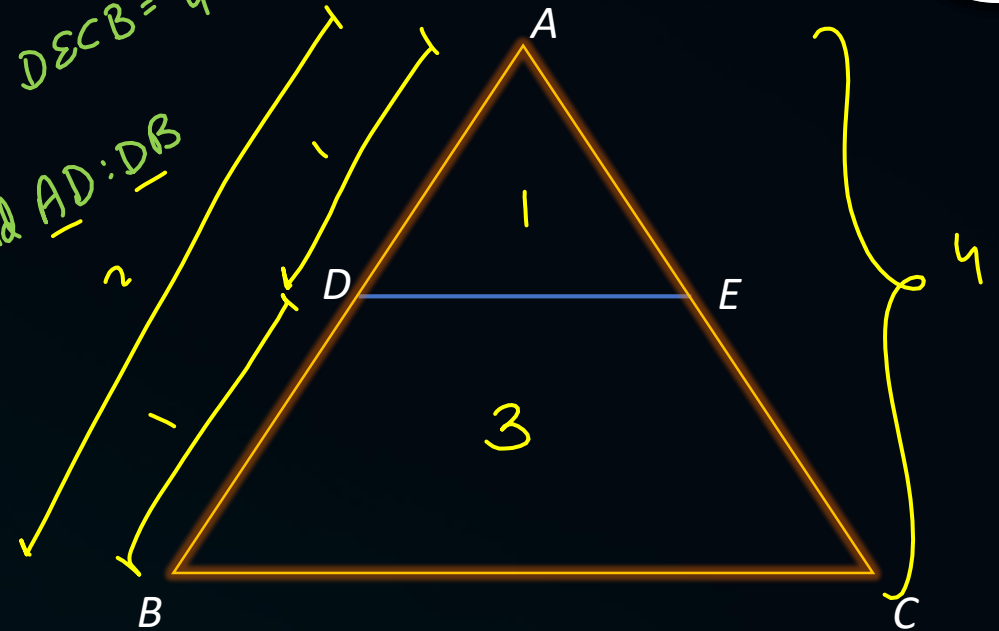
$$\frac{A_{DE}}{A_{BC}} = \frac{8}{27}$$

$$\frac{AE}{AC} = k$$

$$k^2 = \frac{8}{27} = \sqrt{\frac{8}{27}} = \frac{2\sqrt{2}}{3\sqrt{3}}$$

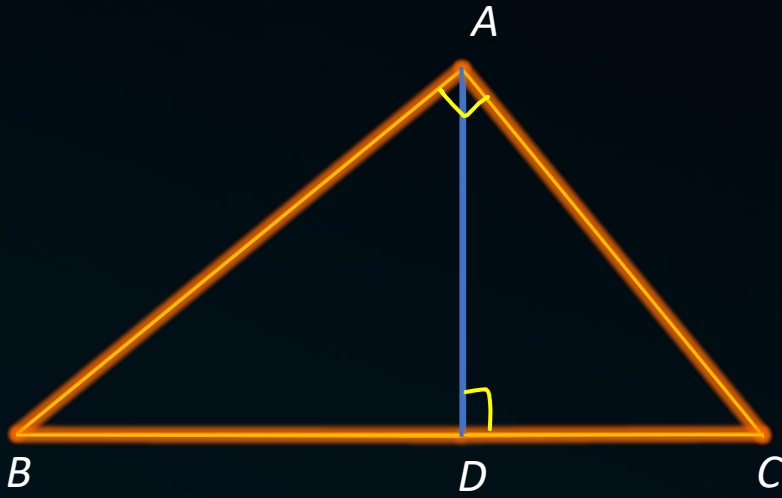
TRAP $DECB = \frac{3}{4} \Delta ABC$

find $\underline{AD}:DB$



$$\frac{AD}{AB} = \frac{1}{2}$$

Important Case - 2



$$AC^2 + AB^2 = BC^2$$

$$(BC)(AD) = (AB)(AC)$$

$$(AD)^2 = (DC)(BD)$$



QUESTION - 4



#Q. Find the value of $x + y$?

- A** 8 cm
- B** 10 cm
- C** 12 cm ✓
- D** 14 cm

$$12 = \frac{36}{3}$$

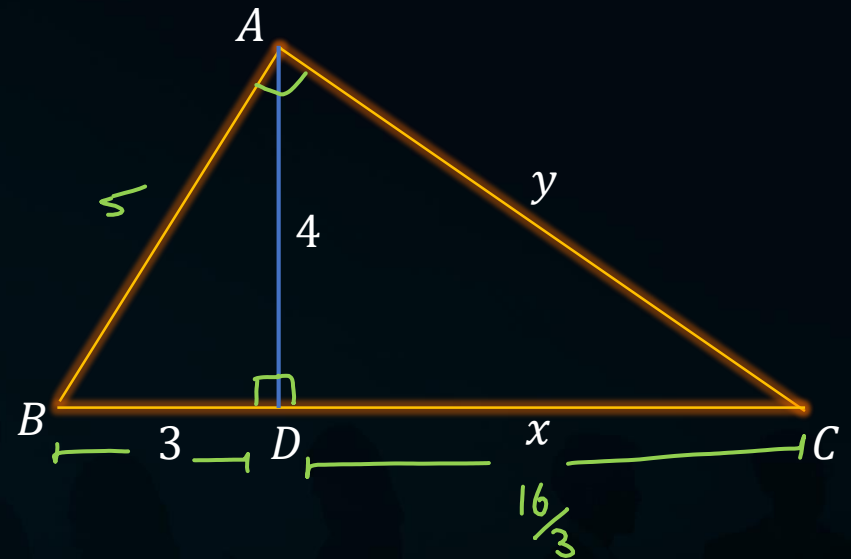
$$16 = 3x$$

$$x = \frac{16}{3}$$

$$5y = \left(3 + \frac{16}{3}\right) 4$$

$$y = \frac{25}{3} (4)$$

$$y = \frac{20}{3}$$

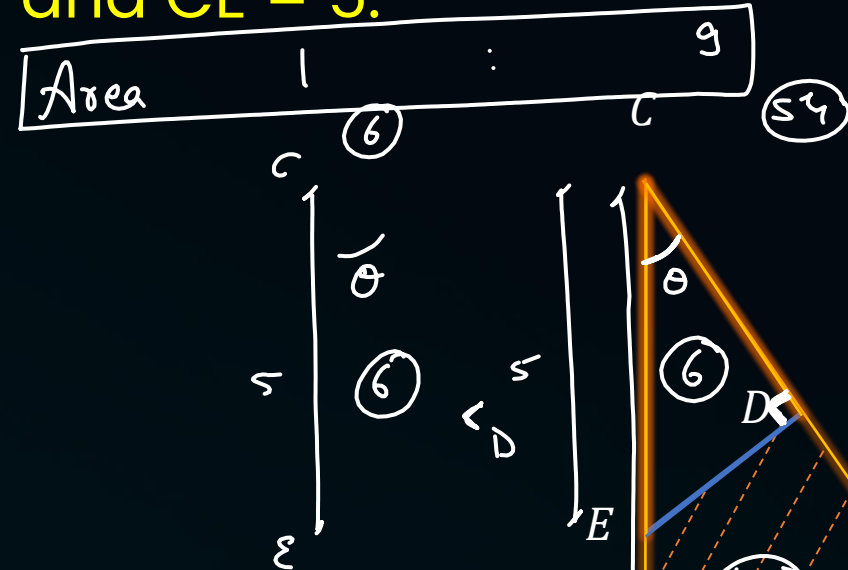
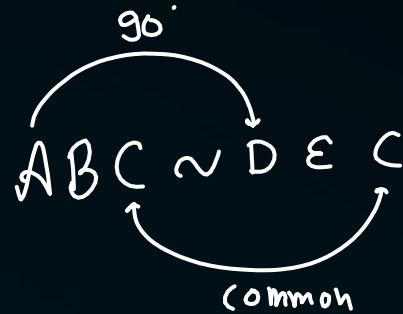


QUESTION - 5

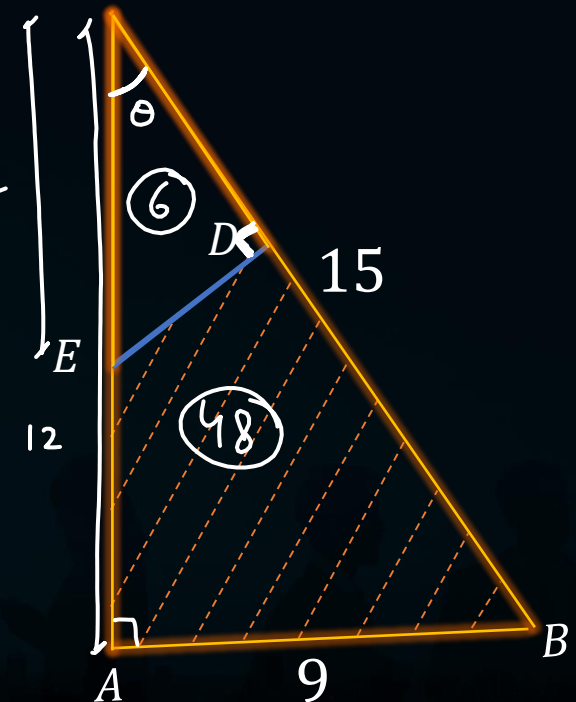


#Q. If ABC is a right angled Triangle Right Angled at A. Find the shaded area if $AB = 9\text{cm}$, $CB = 15\text{cm}$ and $CE = 5$.

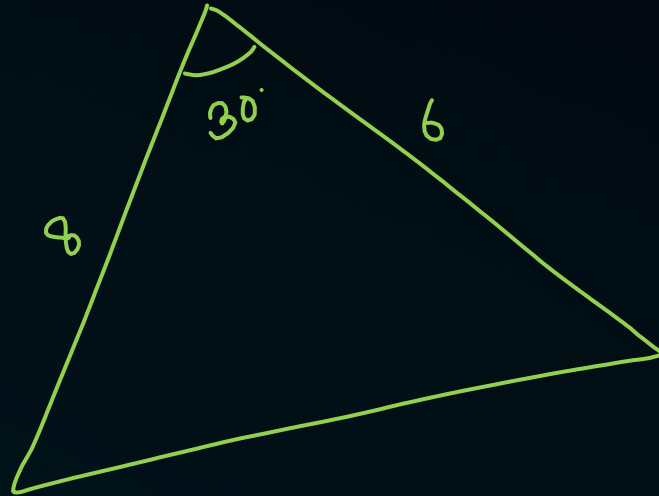
- A** 6 sq.cm
- B** 48 sq.cm ✓
- C** 8 sq.cm
- D** 45 sq.cm



$$A_1(ABC) = \frac{1}{2} (9) (12) = 54$$



Area Last Formula



$$\frac{1}{2} (6)(8) \sin 30^\circ$$

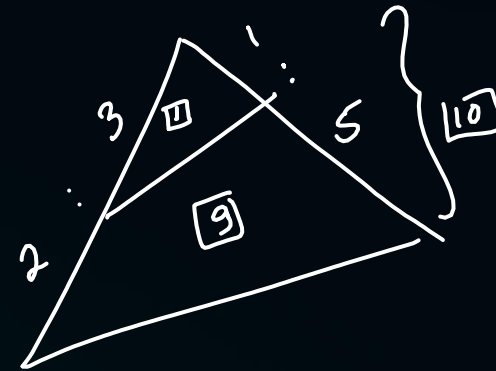
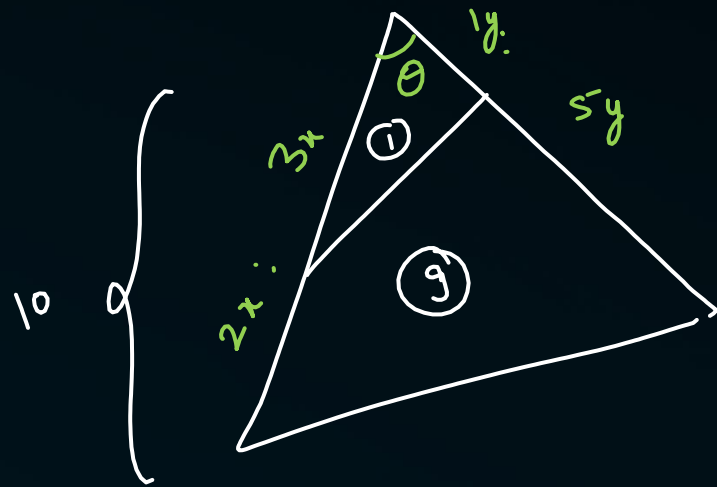
$$\frac{1}{2} (6)(8) \frac{1}{2}$$

$$= 12$$

$$\frac{1}{2} ab \sin \theta$$

θ	$\sin \theta$
0°	$\sqrt{\frac{0}{4}} = 0$
30°	$\sqrt{\frac{1}{4}} = \frac{1}{2}$
45°	$\sqrt{\frac{2}{4}} = \frac{1}{\sqrt{2}}$
60°	$\sqrt{\frac{3}{4}} = \frac{\sqrt{3}}{2}$
90°	$\sqrt{\frac{4}{4}} = 1$

Example - 1

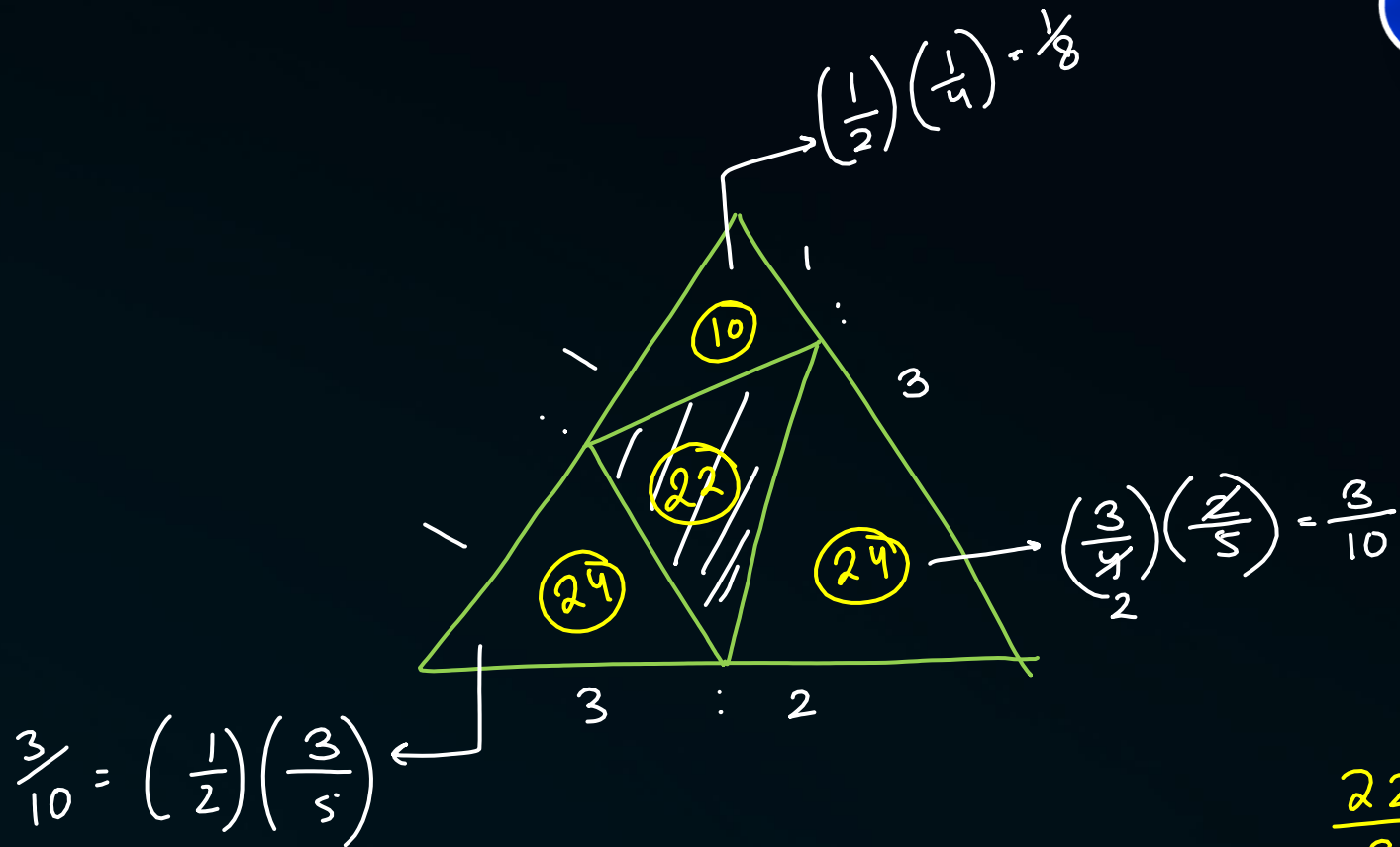


$$\left(\frac{3}{5}\right) \left(\frac{1}{6}\right) = \frac{1}{10}$$

$$\frac{\frac{1}{2} (2x) (y) \sin \theta}{\frac{1}{2} (5x) (\frac{1}{6}y) \sin \theta}$$

$$= \frac{1}{10}$$

Example - 2

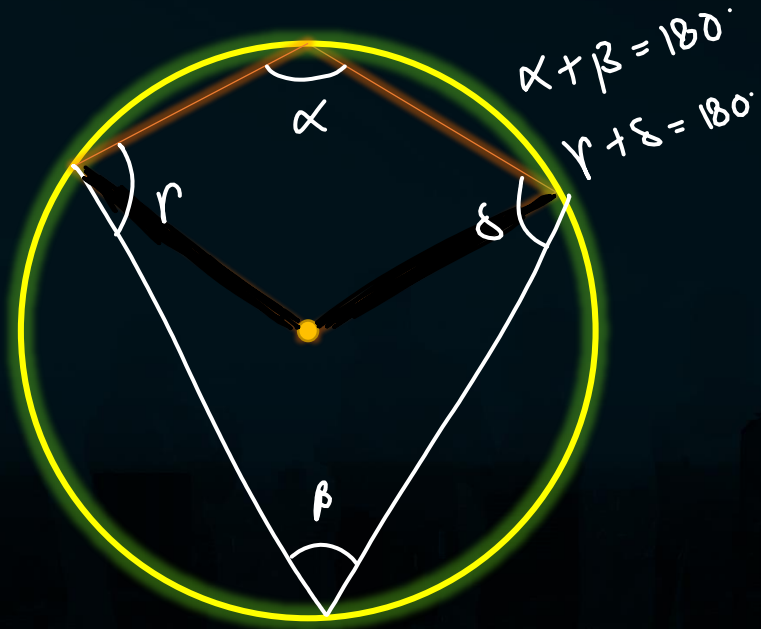
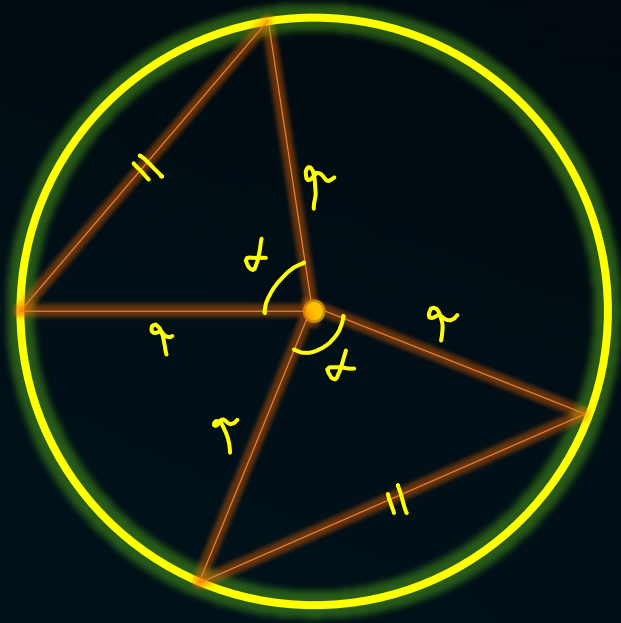


$$\triangle \rightarrow 80$$

$$\frac{22}{80} = \frac{11}{40} \checkmark$$



Topics : Circles



QUESTION - 6



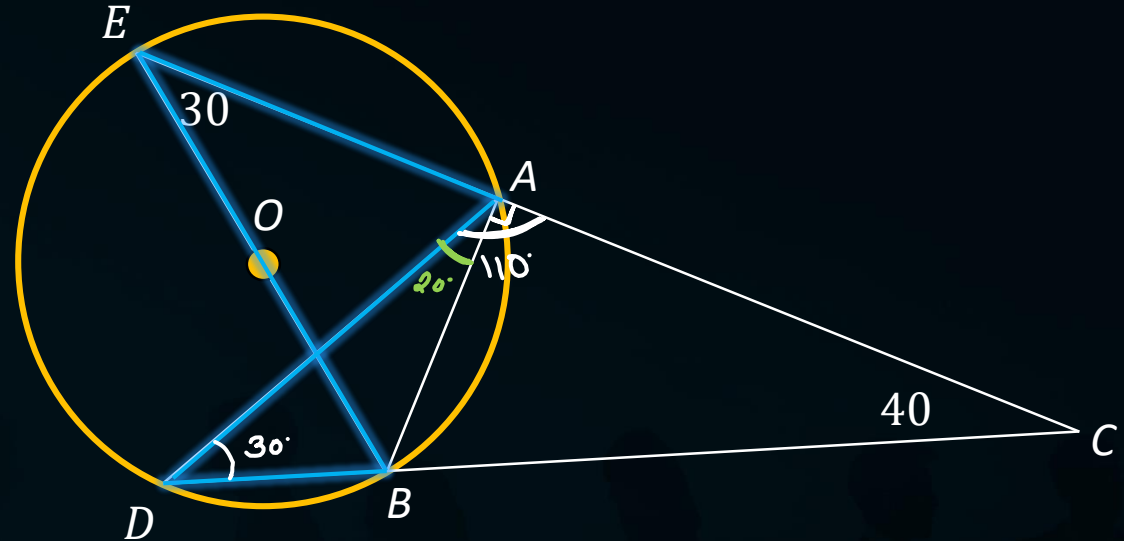
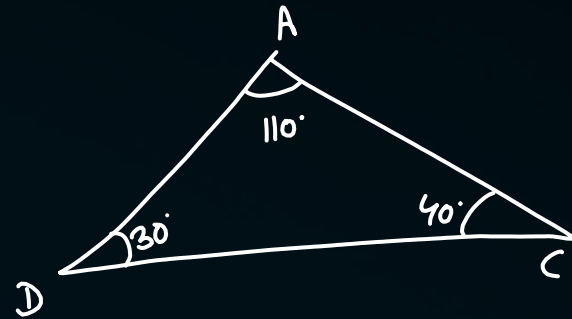
#Q. If BE is the Diameter in the given circle, find Angle DAB.

A 10°

B 20° ✓✓

C 30°

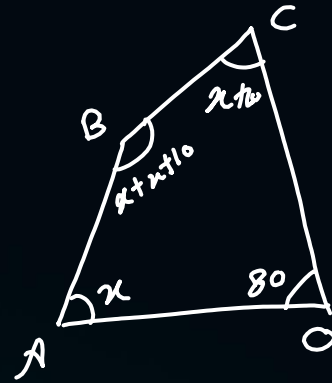
D 40°



QUESTION - 7



#Q. Find the value of x ?



A 50°

B 55°

C 65° ✓✓

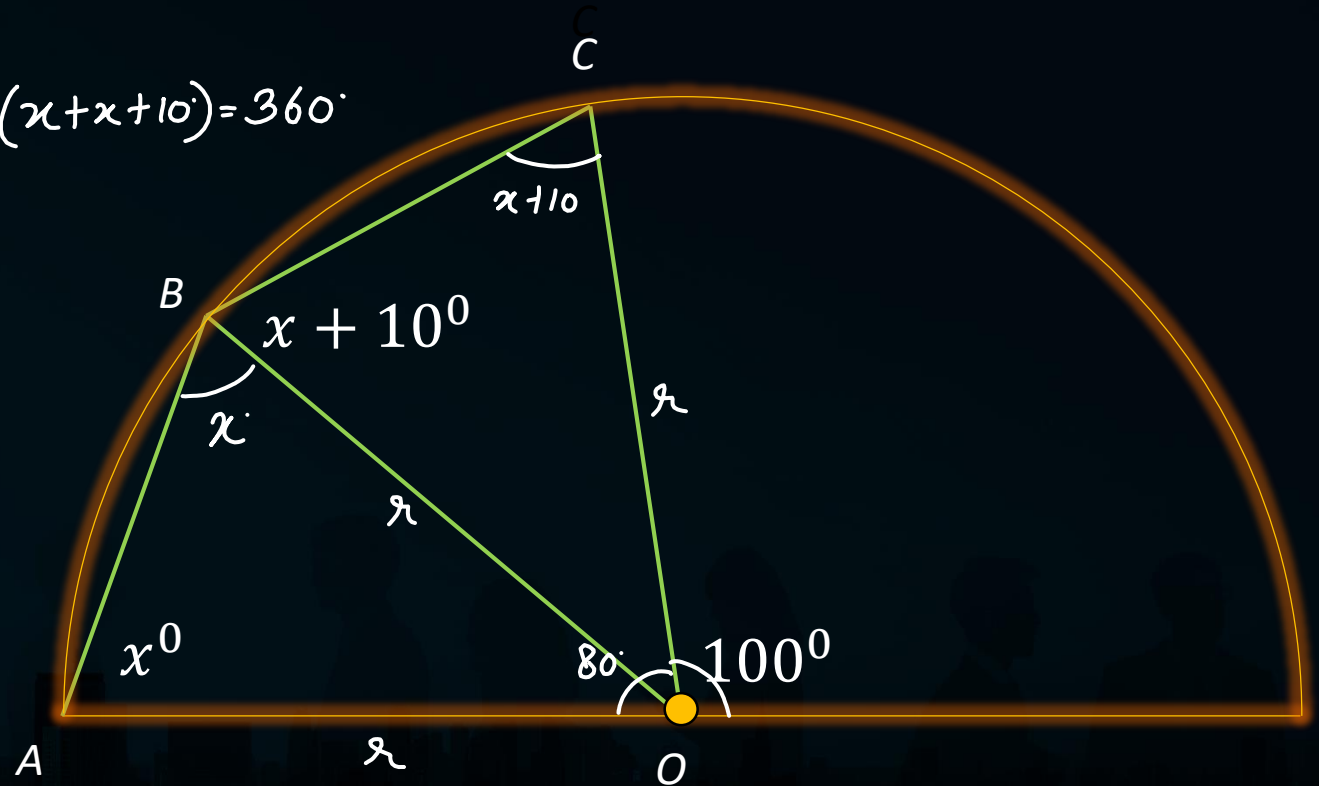
D 110°

$$x^\circ + 80^\circ + x + 10^\circ + (x + x + 10^\circ) = 360^\circ$$

$$4x + 100^\circ = 360$$

$$4x = 260$$

$$x = 65^\circ$$

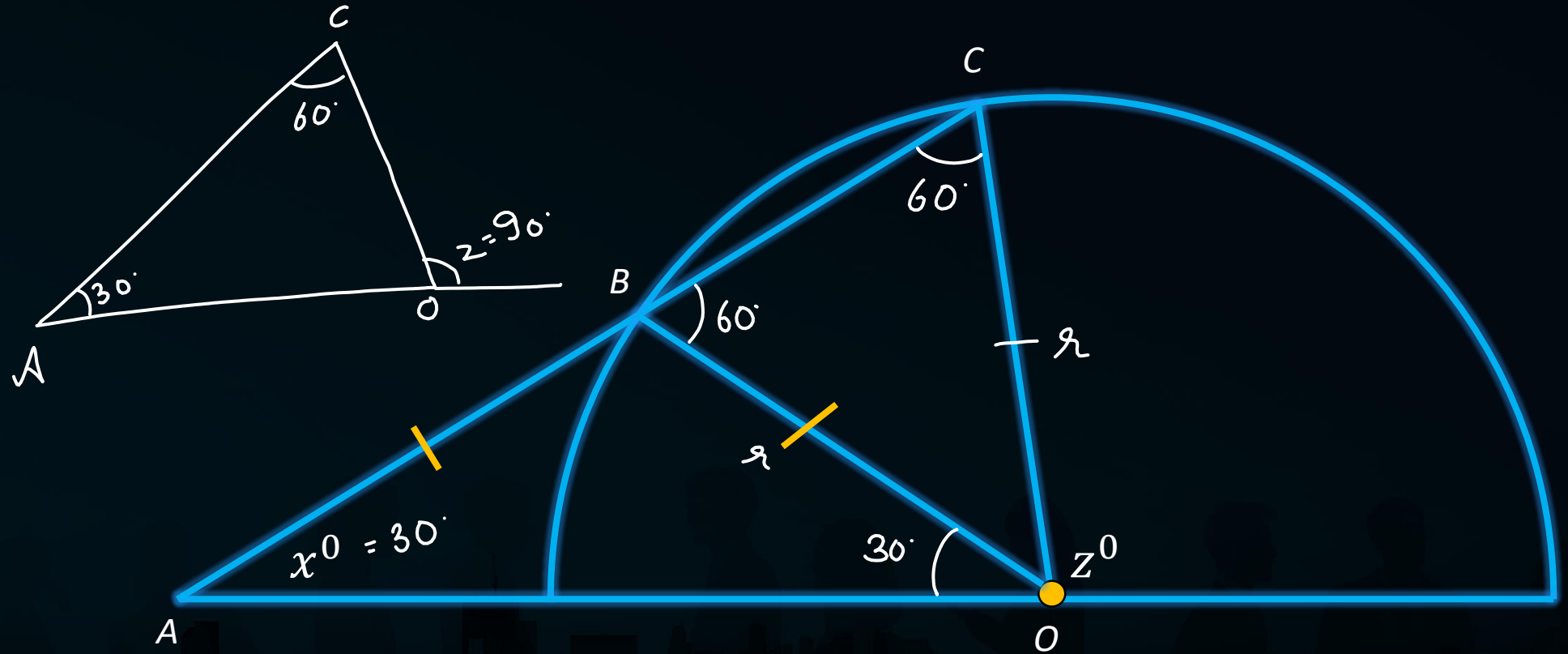


A 50°

B 60°

C 75°

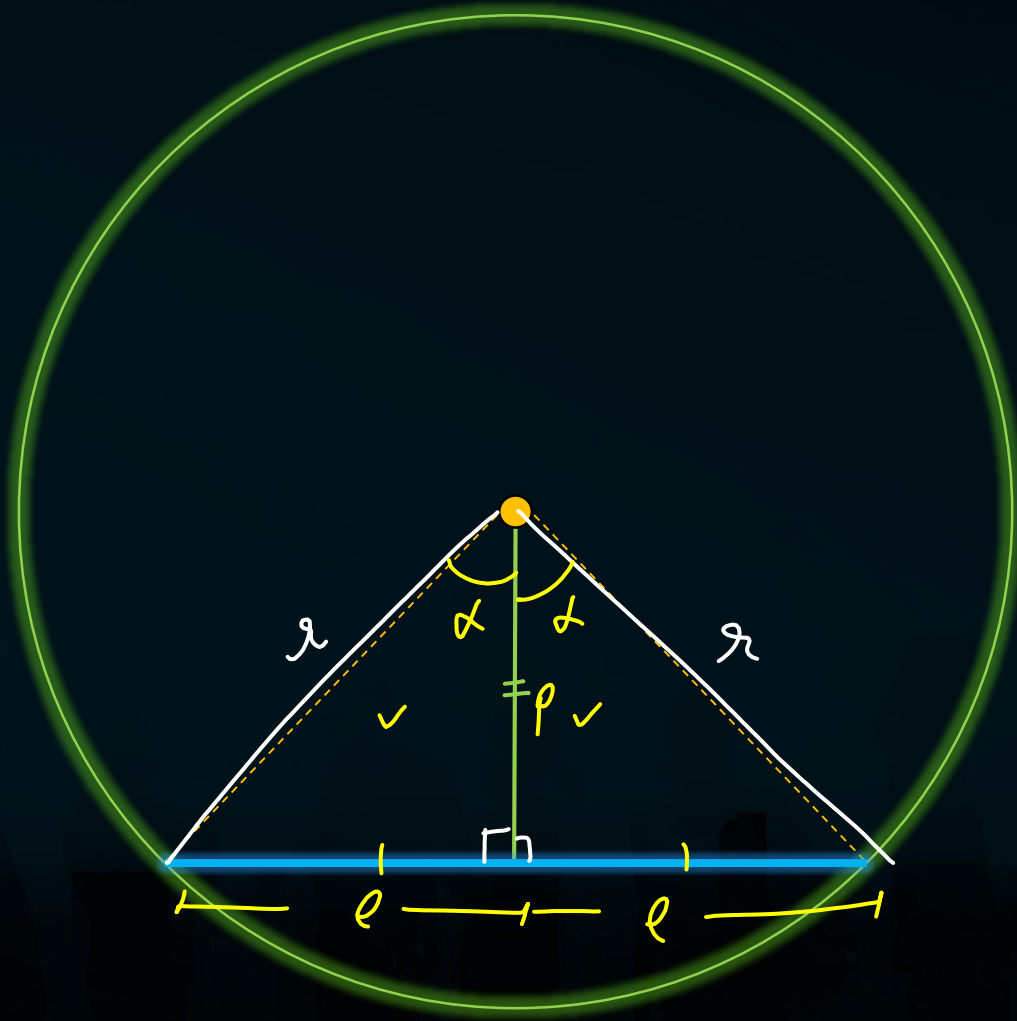
D 90° ✓✓





Length of Chord

Important Case - 1



$$r^2 = p^2 + \ell^2$$

$$\frac{1}{2} (2\ell) (p)$$

$$\ell p = \text{Area}$$

QUESTION - 9



#Q. The distance between two chords of a circle is 14 cm. The length of one chord is 16 cm. What is the length of another chord if the radius is equal to 10 cm.

- A** 16 cm
- B** 20 cm
- C** 12 cm ✓
- D** 14 cm



QUESTION – 10 (CAT 2018 Slot 1)



#Q. In a circle, two parallel chords on the same side of a diameter have lengths 4 cm and 6 cm. If the distance between these chords is 1 cm, then the radius of the circle, in cm, is

A

$\sqrt{13}$

B

$\sqrt{14}$

C

$\sqrt{11}$

D

$\sqrt{12}$

$$4 + 9 = r^2 = 13$$

$$x^2 + 3^2 = r^2$$

$$(x+1)^2 + 2^2 = r^2$$

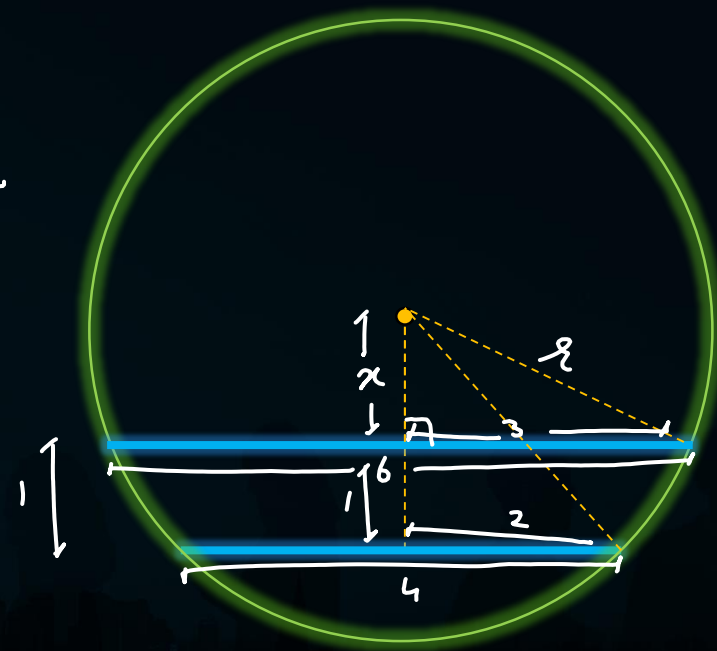
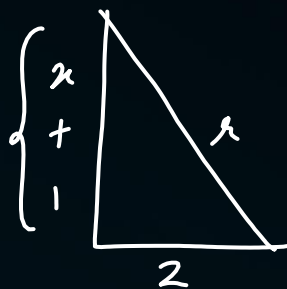
$$(x+1)^2 + 4 = x^2 + 9$$

$$(x+1)^2 - x^2 = 5$$

$$(1)(2x+1) = 5$$

$$2x = 4$$

$$x = 2$$



#Q. Let C_1 and C_2 be concentric circles such that the diameter of C_1 is 2cm longer than that of C_2 . If a chord of C_1 has length 6 cm and is a tangent to C_2 , then the diameter, in cm of C_1 is



SUMMARY



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ADVANCE TRIANGLES



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